

GRAPH COLORING

Introduction

- problem of determining the least number of colors that can be used to color a map so that adjacent regions never have the same color.
- for the map shown on the left in Figure 1, four colors suffice, but three colors are not enough.
- In the map on the right in Figure 1, three colors are sufficient (but two are not).

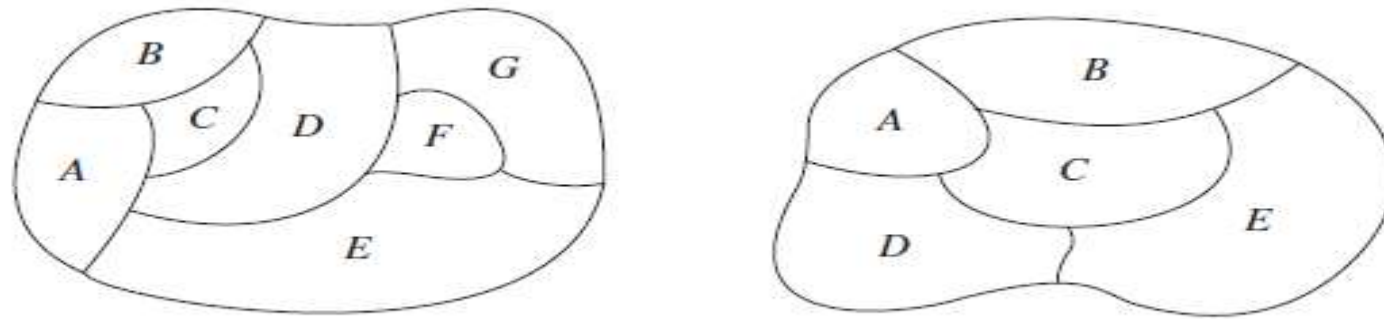


FIGURE 1 Two Maps.

- Each map in the plane can be represented by a graph.
- Each region of the map is represented by a vertex
- Two regions that touch at only one point are not considered adjacent.
- The resulting graph is called the **dual graph** of the map.
- Figure 2 displays the dual graphs that correspond to the maps shown in Figure 1.

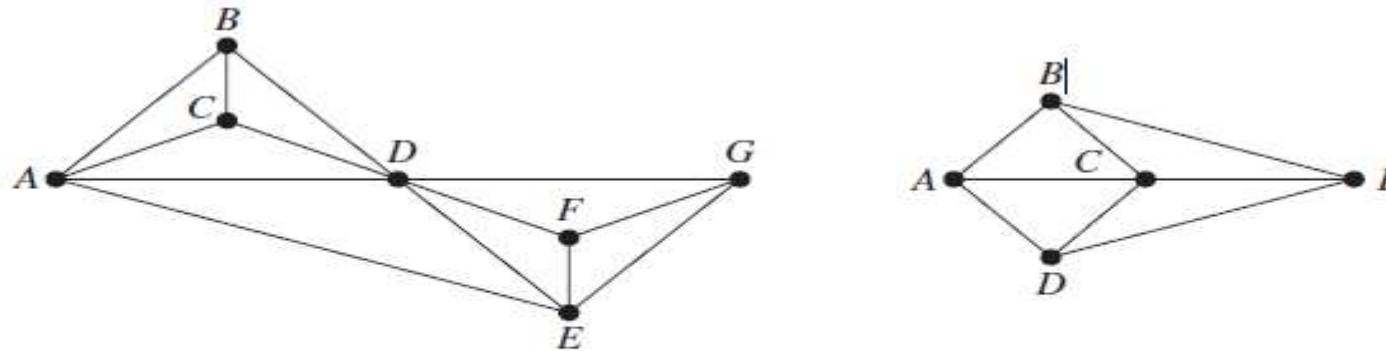


FIGURE 2 Dual Graphs of the Maps in Figure 1.

- The problem of coloring the regions of a map is equivalent to the problem of coloring the vertices of the dual graph so that no two adjacent vertices in this graph have the same color.

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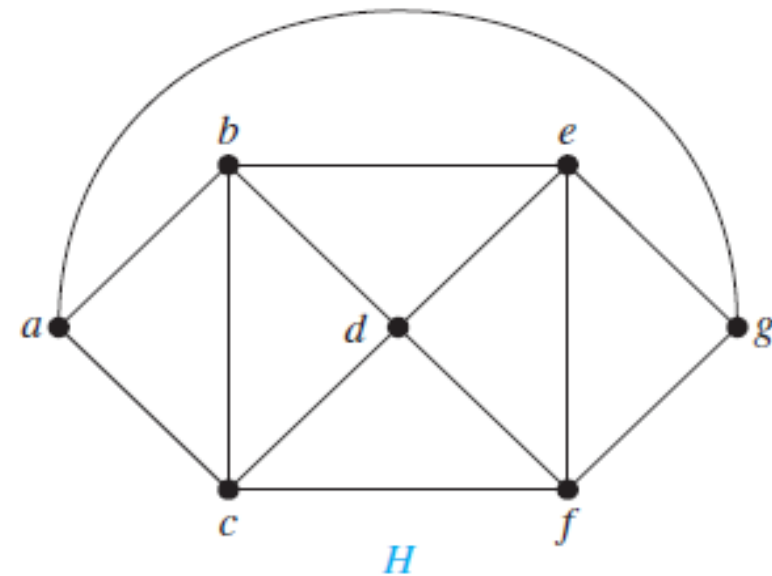
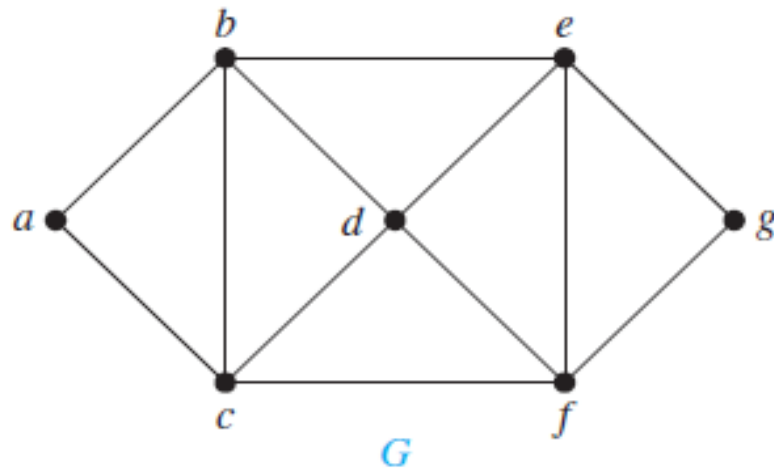
- A *coloring* of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.
- A graph can be colored by assigning a different color to each of its vertices.
- The *chromatic number* of a graph is the least number of colors needed for a coloring of this graph. The chromatic number of a graph G is denoted by $\chi(G)$. (Here χ is the Greek letter *chi*.)
- The chromatic number of a planar graph is the same as asking for the minimum number of colors required to color a planar map so that no two adjacent regions are assigned the same color.

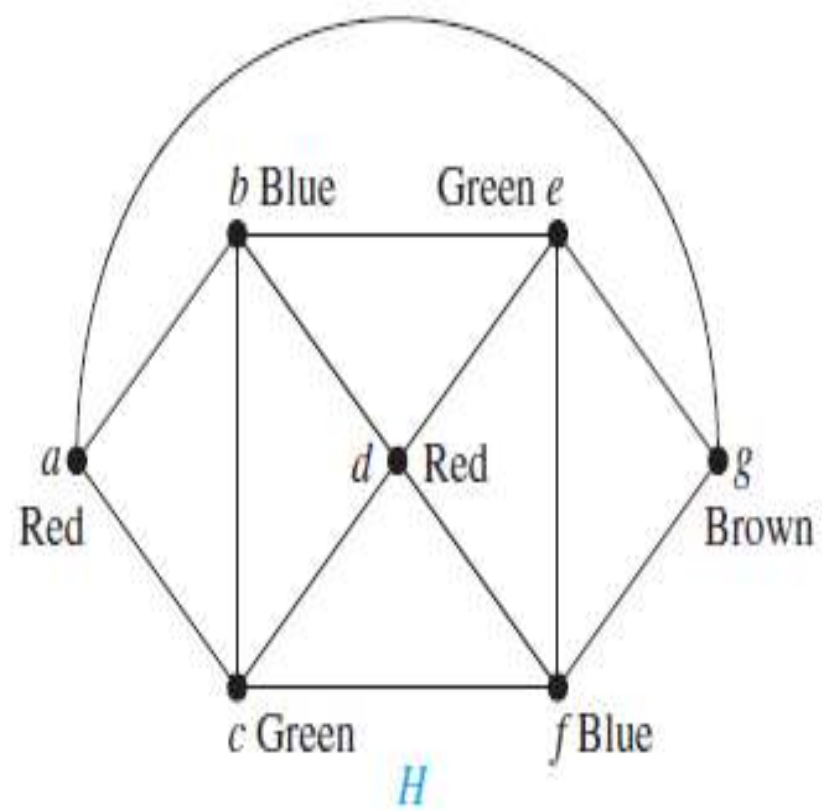
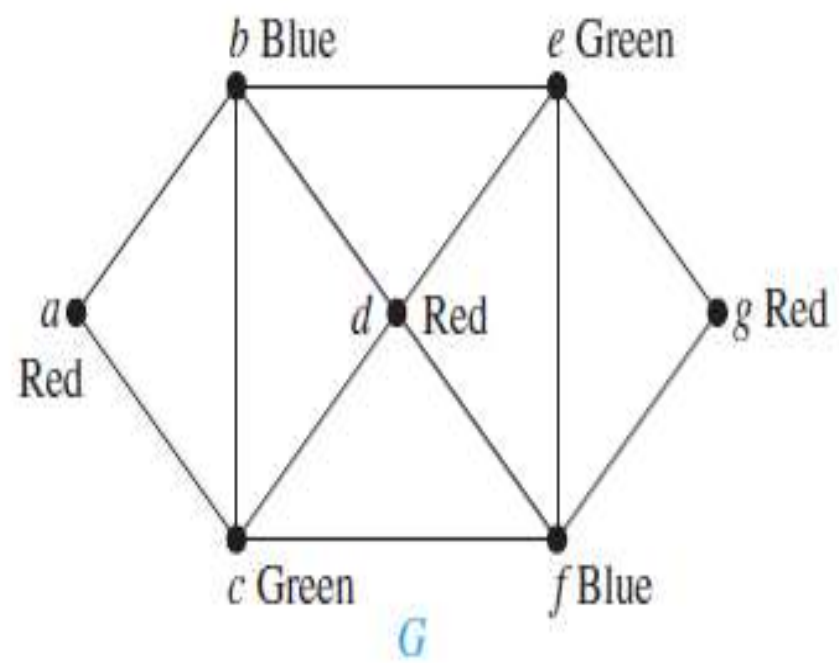
THEOREM 1

THE FOUR COLOR THEOREM

The chromatic number of a planar graph is no greater than four.

- What are the chromatic numbers of the graphs G and H shown in Figure?





- The chromatic number of G is at least three, because the vertices a , b , and c must be assigned different colors. To see if G can be colored with three colors, assign red to a , blue to b , and green to c . Then, d can (and must) be colored red because it is adjacent to b and c .
- Furthermore, e can (and must) be colored green because it is adjacent only to vertices colored red and blue, and f can (and must) be colored blue because it is adjacent only to vertices colored red and green. Finally, g can (and must) be colored red because it is adjacent only to vertices colored blue and green. This produces a coloring of G using exactly three colors.
- Figure 4 displays such a coloring.
- The graph H is made up of the graph G with an edge connecting a and g . Any attempt to color H using three colors must follow the same reasoning as that used to color G , except at the last stage, when all vertices other than g have been colored. Then, because g is adjacent (in H) to vertices colored red, blue, and green, a fourth color, say brown, needs to be used. Hence, H has a chromatic number equal to 4. A coloring of H is shown in Figure 4.

What is the chromatic number of K_n ?

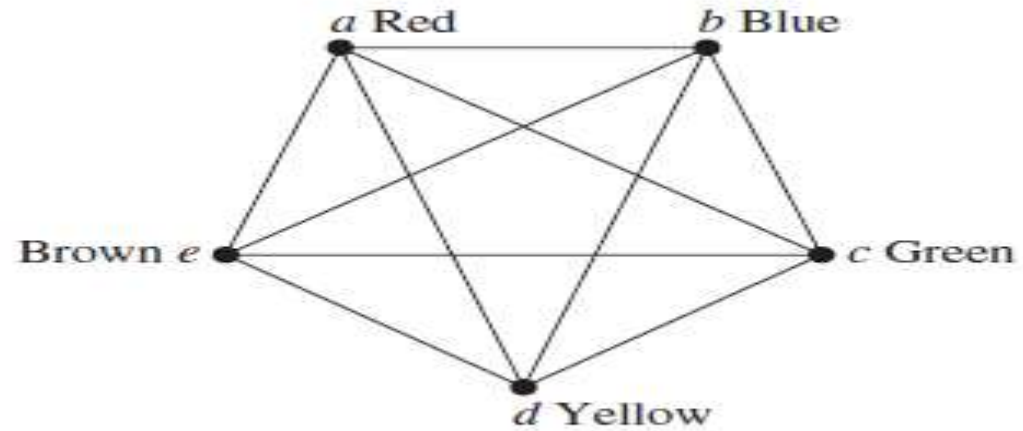


FIGURE 5 A Coloring of K_5 .

- A coloring of K_n can be constructed using n colors by assigning a different color to each vertex.
- No two vertices can be assigned the same color, because every two vertices of this graph are adjacent.
- Hence, the chromatic number of K_n is n . That is, $\chi(K_n) = n$.
- A coloring of K_5 using five colors is shown in Figure 5.

Example: What is the chromatic number of the complete bipartite graph $K_{m,n}$, where m and n are positive integers?

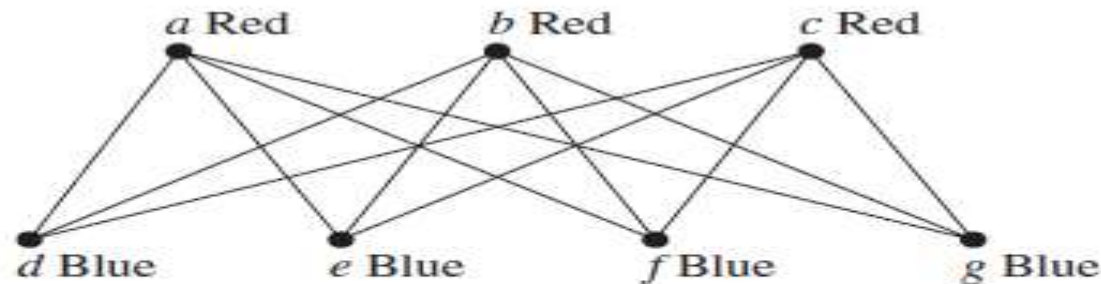


FIGURE 6 A Coloring of $K_{3,4}$.

- The number of colors needed may seem to depend on m and n .
- only two colors are needed, because $K_{m,n}$ is a bipartite graph. Hence, $\chi(K_{m,n}) = 2$.
- This means that we can color the set of m vertices with one color and the set of n vertices with a second color. Because edges connect only a vertex from the set of m vertices and a vertex from the set of n vertices, no two adjacent vertices have the same color.
- A coloring of $K_{3,4}$ with two colors is displayed in Figure 6.
- Example: What is the chromatic number of the graph C_n , where $n \geq 3$? (Recall that C_n is the cycle with n vertices.)

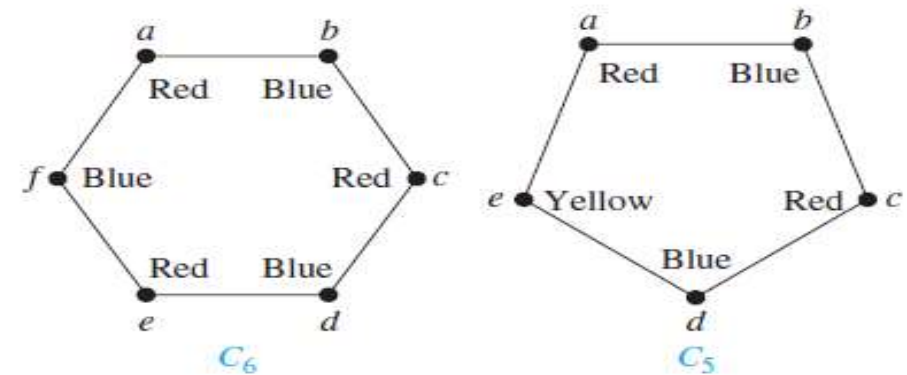


FIGURE 7 Colorings of C_5 and C_6 .

- let $n = 6$. Pick a vertex and color it red. Proceed clockwise in the planar depiction of C_6 shown in Figure 7. It is necessary to assign a second color, say blue, to the next vertex reached. Continue in the clockwise direction; the third vertex can be colored red, the fourth vertex blue, and the fifth vertex red. Finally, the sixth vertex, which is adjacent to the first, can be colored blue. Hence, the chromatic number of C_6 is 2. Figure 7 displays the coloring constructed here.
- Next, let $n = 5$ and consider C_5 . Pick a vertex and color it red. Proceeding clockwise, it is necessary to assign a second color, say blue, to the next vertex reached. Continuing in the clockwise direction, the third vertex can be colored red, and the fourth vertex can be colored blue. The fifth vertex cannot be colored either red or blue, because it is adjacent to the fourth vertex and the first vertex. Consequently, a third color is required for this vertex. Note that we would have also needed three colors if we had colored vertices in the counterclockwise direction. Thus, the chromatic number of C_5 is 3. A coloring of C_5 using three colors is displayed in Figure 7.
- $\chi(C_n) = 2$ if n is an even positive integer with $n \geq 4$ and $\chi(C_n) = 3$ if n is an odd positive integer with $n \geq 3$.

Application of Graph Coloring

- **Scheduling Final Exams** How can the final exams at a university be scheduled so that no student has two exams at the same time?

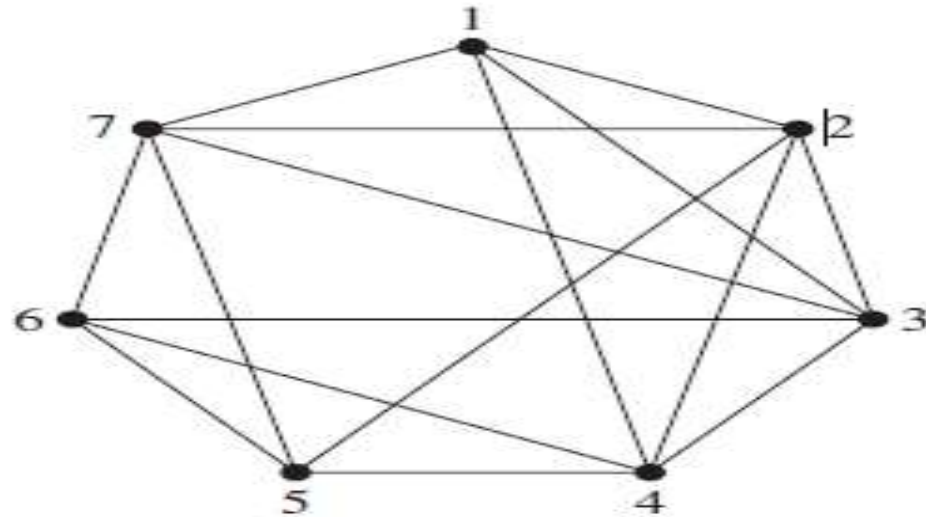


FIGURE 8 The Graph Representing the Scheduling of Final Exams.

- This scheduling problem can be solved using a graph model, with vertices representing
- courses and with an edge between two vertices if there is a common student in the courses they represent.
- Each time slot for a final exam is represented by a different color.
- A scheduling of the exams corresponds to a coloring of the associated graph.
- For instance, suppose there are seven finals to be scheduled. Suppose the courses are numbered 1 through 7.
- Suppose that the following pairs of courses have common students: 1 and 2, 1 and 3, 1 and 4, 1 and 7, 2 and 3, 2 and 4, 2 and 5, 2 and 7, 3 and 4, 3 and 6, 3 and 7, 4 and 5, 4 and 6, 5 and 6, 5 and 7, and 6 and 7.
- In Figure 8 the graph associated with this set of classes is shown.
- A scheduling consists of a coloring of this graph.
- Because the chromatic number of this graph is 4, four time slots are needed. A coloring of the graph using four colors and the associated schedule are shown in Figure 9.

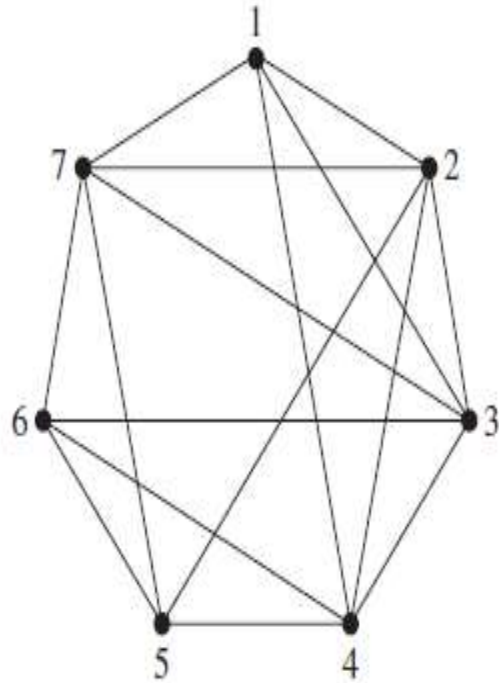
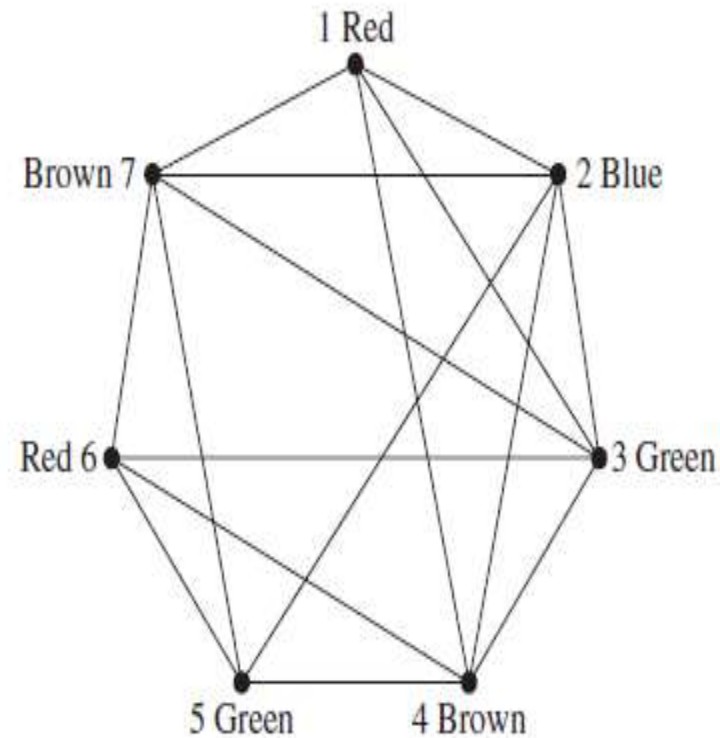


FIGURE 8 The Graph
Representing the Scheduling
of Final Exams.



Time Period	Courses
I	1, 6
II	2
III	3, 5
IV	4, 7

FIGURE 9 Using a Coloring to Schedule Final Exams.